

Lique Retentra WS-50

High Performance, Modified Polycarboxylic Ether based Hyperplasticiser

Lique Retentra WS-50 is a new generation hyperplasticiser based on modified polycarboxylic ether (PCE) technology. It is engineered to enhance cement dispersion significantly, reducing the water required for mixing while maintaining superior flow.

The mechanism involves the adsorption of PCE onto cement granules to create electrostatic dispersion, combined with side-chain polymer backbones that generate **steric hindrance**. This dual-action physical and electrostatic barrier results in highly flowable concrete with long-term dispensability and a major reduction in water content.

Uses:

- Production of high-grade strength class concrete.
- Creating dense, rheodynamic concrete with high ultimate strength.
- Applications requiring high flowability and extended working times.
- Suitable for Self-Compacting Concrete (SCC) and Smart Dynamic Concrete (SDC) mixes.

Technical Properties:

Property	Specifications
Appearance	Light Yellow Liquid
Specific Gravity	1.10 $\pm$ 0.04
pH Value	7 $\pm$ 1
Chloride Content	NIL (Chloride-free)
Air Entrainment	$\leq$ 1.5% marginal increase
Compatibility	Compatible with OPC, Slag cements, Microsilica, Fly Ash, and GGBFS
Standard Compliance	Conforms to IS:9103-1999 and ASTM C494 Type F & G
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Appearance	Light Yellow Liquid
Specific Gravity	1.10 $\pm$ 0.04
pH Value	7 $\pm$ 1

Standards Compliance

Lique Retentra WS-50 conforms to the following international standards:

- ASTM C494: Type F & G
- IS:9103-1999: (Reaffirmed 2004)

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Key Advantages

- **Enhanced Workability:** Significant improvement in flow without adding extra water.
- **Pumping Efficiency:** Reduces internal friction and thixotropy, minimizing segregation risks during pumping.
- **Durability:** Minimizes shrinkage cracking to produce impermeable, close-textured concrete.
- **Corrosion Protection:** Reduced permeability protects embedded steel from water migration.
- **Superior Bond:** Increases the bond strength between concrete and reinforcement steel.

Application Guidelines

Dosage

The optimum dosage must be determined via site trials with specific mix designs.

- **General Range:** 0.4% to 1.2% by weight of cement.
- **For High Flow/SCC:** 0.6% to 1.0% by weight of cement.

Method of Use

- **Ready to Use:** Lique Retentra WS-50 should be added directly into the mixer along with the mixing water.
- **Enhanced Performance:** For best results, add the admixture to "moist" concrete after 60-70% of the mixing water has been introduced.
- **Constraint: Never** add to dry cement or dry aggregates.

Packaging & Storage

- **Packaging:** Supplied in 230kg drums or bulk tankers.
- **Shelf Life:** 12 months from the date of manufacture when stored in original, unopened containers.
- **Storage:** Keep in shaded areas between 20-35c

Health & Safety

- **Safety:** Non-toxic and non-flammable.
 - **Skin Contact:** Wash splashes with plenty of water.
 - **Eye Contact:** Flush immediately with clean water and seek medical advice.
- Disclaimer:** While technical details are based on our best knowledge, they are intended as indications only. Practical site trials are mandatory to determine exact dosages. The user is responsible for the final consequences of product use.